Patient Drug Review Classification: A Multi-Classifier NLP Comparison on Condition Identification from Drug Reviews

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Abstract

Patient-written drug reviews are largely untapped sources of real-world clinical information, capturing how people describe their treatment experiences in everyday language. Most prior work on this data has focused on sentiment or side effects, with less attention paid to identifying the actual condition being treated. In this study, we compare three different NLP approaches for classifying medical conditions from review text: a TF-IDF-based Support Vector Machine (SVM), a fine-tuned BERT model, and a zero-shot large language model (LLM). Using the UCI Drug Review dataset, we focus on six common conditions, birth control, depression, anxiety, acne, ADHD, and type 2 diabetes, to ensure sufficient data while still reflecting real-world class imbalance. All models are evaluated on the same held-out test set using macro-averaged precision, recall, and F1-score. The SVM performs best overall (macro F1 = 0.96), with BERT close behind (0.94). The zero-shot LLM, despite requiring no training data, achieves a competitive F1 of 0.91 when paired with a well-designed prompt. Across all models, conditions with distinctive vocabulary (such as birth control and diabetes) are consistently easy to classify, while anxiety and depression remain difficult to separate due to overlapping language. We also find that prompt design has a large impact in the zero-shot setting, improving performance by 14 F1 points more than the difference between SVM and BERT. Overall, our results suggest that simple classical methods remain very strong baselines for this task, while zero-shot LLMs provide a practical alternative when labeled data or compute resources are limited.

1. Introduction and background

Patients increasingly document their medication experiences online, producing a large and growing corpus of informal but information-rich drug reviews. These texts capture real-world treatment outcomes in everyday language, making them a valuable complement to clinical trial data for understanding how drugs perform across diverse patient populations. The scale and unstructured nature of this data, however, make manual review impractical, and automated NLP methods have been widely explored as a means of extracting clinically useful signals from free-text sources (Topaz et al., 2016). Research in this space has grown considerably, with applications ranging from pharmacovigilance and adverse event detection to drug recommendation systems and population health surveillance (Sarker et al., 2015). Despite this progress, much of the existing work treats patient reviews primarily as a source of sentiment or opinion rather than as structured medical information. Studies tend to ask whether a patient rated a drug positively or negatively or which side effects are mentioned most frequently, rather than the condition the patient was being treated for (Grasser et al., 2018). This distinction matters: knowing the reported condition is essential for downstream tasks like condition-specific efficacy comparisons or targeted pharmacovigilance signals, yet it has received comparatively little attention as a standalone classification objective.

Research Gap

To our knowledge, no published study has placed condition classification from patient drugs. reviews at the center of a systematic, multi-architecture benchmark. The UCI Drug Review corpus (Kallumadi & Grer, 2018) has been used in prior work, but typically for sentiment or rating prediction rather than condition identification. Classical sparse-text methods such as SVMs (Cortes & Vapnik, 1995) and contextual transformer models such as BERT (Devlin et al., 2019) have each been applied to biomedical text tasks in isolation, but their relative strengths on this particular problem have not been directly compared. The recent emergence of large language models capable of zero-shot classification without any task-specific training (Brown et al., 2020) adds a third paradigm whose practical value on this task is entirely unexplored. It is also unclear how the highly imbalanced, long-tail label space of 916 conditions affects each approach differently.

Project Objectives

This project fills that gap by running a controlled, head-to-head comparison of three classifier families on the same dataset and evaluation protocol. We select six high-frequency conditions from the UCI Drug Review corpus to ensure sufficient labeled examples for supervised training while preserving meaningful class imbalance. A TF-IDF SVM serves as the classical baseline, a fine-tuned BERT model represents supervised deep learning, and a zero-shot LLM represents the prompt-only paradigm. All three are evaluated on an identical held-out test set using macro-averaged precision, recall, and F1-score to give equal weight to every condition regardless of its frequency. Our aim is not only to rank the classifiers but also to understand where and why each approach succeeds or fails and what that reveals about the underlying structure of the classification problem.

2. Data preparation

The corpus, made publicly available through the UCI Machine Learning Repository, comprises 215 063 patient-written reviews describing experiences with 3654 drugs and 916 conditions. Each review record contains a unique identifier, the drug name, the reported condition, a free-text comment, the date, a 1–10 star rating, and a useful-count field; these attributes together capture both quantitative and qualitative aspects of patient feedback. Since the comments are provided by consumers, the recorded "condition" is a coarse indication rather than a formal diagnosis.

For modeling, the dataset is supplied in separate training and test files. When the two splits are combined, there are 161 297 training entries and 53 766 test entries, which is about a 3:1 ratio. We verified that unique identifiers do not overlap between the splits, ensuring independence for evaluation. No comment texts are missing, but approximately 1194 records lack a condition label and were excluded from supervised learning.

Exploratory analysis reveals that the label space is highly skewed. In total there are 916 distinct condition classes, yet only a handful of conditions dominate the sample: the largest class contains 38,436 reviews, while many rare conditions appear only once. The typical review, after basic cleaning, contains about 84 words (≈443 characters), but length distributions are right-skewed, with a long tail of exceptionally long comments. Figure 1 illustrates this long-tail

phenomenon: most bars, representing conditions, are barely visible compared with the few towering, frequent categories. Such imbalance implies that accuracy alone would favor the dominant classes; thus, later modeling uses various metrics to reflect performance across all conditions.

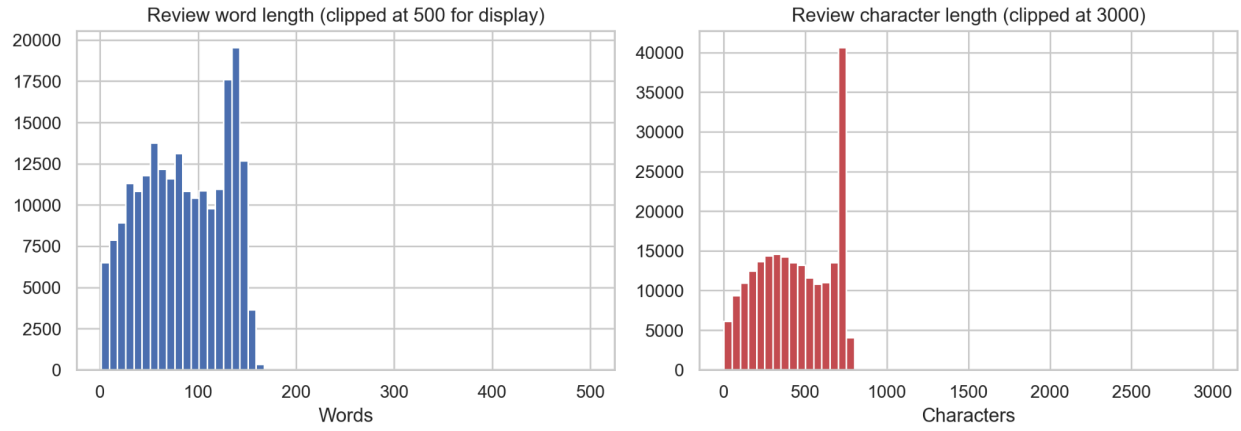


Figure 1: Review length distribution

Besides the outcome label, the data contain the drug name and a user rating; in this study, only the free-text review was used as model input, and the drug-name and rating fields were excluded as features. Preprocessing is necessary to transform the free-text reviews into numerical features. Following Zipf's law, a small set of high-frequency function words accounts for much of the corpus. We lowercase all text, remove punctuation and stop words, and apply stemming or lemmatization to normalize word forms. These steps reduce noise and prepare the text for feature extraction via bag-of-words, TF–IDF, or neural embeddings.

3. Methods

Method 1: SVM + TF-IDF Baseline

To establish a strong classical machine learning baseline for classifying patients' conditions, we applied a Support Vector Machine (SVM) classifier combined with Term Frequency–Inverse Document Frequency (TF-IDF) feature representation. SVM was selected because it has been shown to perform exceptionally well on text classification tasks, particularly when the input is a high-dimensional, sparse vectorized representation (Cortes & Vapnik, 1995). This approach contrasts with our transformer-based models, which rely on contextual embeddings rather than sparse term-frequency features.

Selection of Target Conditions

The UCI Drug Review dataset contains 916 unique conditions, the vast majority of which appear in only a handful of reviews and are unsuitable for supervised classification. To address this severe class imbalance, we restricted the SVM task to six high-frequency clinical conditions that retained sufficient samples in both the training and test splits after preprocessing: birth control, depression, anxiety, acne, ADHD, and type 2 diabetes. After filtering, the training subset comprised 55,286 reviews, and the test set comprised 18,432 reviews. The class distribution remained substantially skewed toward birth control, which alone accounted for roughly 52% of the training data; diabetes type 2 was the smallest retained class, with 2,554 training and 808 test reviews.

Text Processing and Feature Generation

Reviews were lowercased and converted into numerical features via TF-IDF. Both unigrams and bigrams (ngram_range=(1,2)) were extracted so that the feature space captured not only individual tokens (e.g., "focus" and “skin”) but also compound expressions such as “birth control” and “panic attack.” English stop words were removed, and the vocabulary was capped at the 30,000 most informative tokens, yielding a sparse feature matrix of shape (55286, 30000) for training and (18432, 30000) for testing. TF-IDF is well suited for this task because it down-weights highly frequent terms and emphasizes tokens that are distinctive to specific medical conditions.

Multi-Class SVM Framework

Because this is a multi-class problem, we evaluated two standard decomposition strategies. Under One-vs-Rest (OvR), one binary classifier is trained per condition against all others. Under One-vs-One (OvO), one classifier is trained for every pair of conditions, with predictions resolved by majority vote. Both strategies use LinearSVC with a regularization parameter. C=1.0. Comparing the two decompositions allows us to assess whether finer-grained pairwise decision boundaries provide any advantage over class-vs-rest boundaries on this dataset.

Method 2: Fine-Tuned BERT

As the second classifier, we fine-tuned a pre-trained BERT model (Devlin et al., 2019) on the same six-condition drug review dataset used for the SVM baseline. BERT (Bidirectional Encoder Representations from Transformers) reads each token in the context of all surrounding tokens simultaneously, enabling it to capture contextual meaning that bag-of-words representations cannot. Whereas the SVM assigns a fixed weight to each term regardless of its surrounding context, BERT can distinguish "not anxious" from "anxious" and "I use this for anxiety, not depression" from a review where those same words appear in the opposite framing. This contextual sensitivity is particularly relevant for the anxiety/depression boundary, which is the dominant error pattern in the SVM.

Model Architecture

We used bert-base-uncased, the standard 12-layer, 110-million-parameter variant of BERT pre-trained on English Wikipedia and BooksCorpus. The "uncased" variant treats uppercase and lowercase letters identically, which is appropriate for patient drug reviews where capitalization is inconsistent. A linear classification head mapping the [CLS] token representation to six output logits was added on top of the pre-trained encoder, and the entire model was fine-tuned end-to-end.

Training Setup

Input sequences were tokenized using the bert-base-uncased WordPiece tokenizer and truncated or padded to a maximum length of 128 tokens. This length covers the vast majority of drug reviews without incurring the quadratic memory cost of longer sequences. Training used a batch size of 32, three epochs, and the AdamW optimizer (Loshchilov & Hutter, 2019) with a learning rate of 2e-5 and weight decay of 0.01. A linear warmup schedule increased the learning rate over the first 10% of training steps before decaying linearly to zero, a standard strategy for stabilizing early BERT fine-tuning. All experiments were run on a single NVIDIA T4 GPU.

Handling Class Imbalance

To address the 11:1 class imbalance between Birth Control (28,788 training reviews) and Diabetes, Type 2 (2,554 training reviews), we computed inverse-frequency class weights and

passed them as a weight vector to the cross-entropy loss function. This penalizes misclassifications of minority classes more heavily during backpropagation, equivalent to the class_weight='balanced' strategy used in the SVM, and avoids discarding any training data. The resulting class weights ranged from 0.32 for birth control to 3.61 for diabetes, type 2.

Method 3: Zero-Shot LLM

As a third classifier, we evaluated a zero-shot large language model (LLM) approach using the Anthropic Claude API. Unlike the SVM and BERT models, zero-shot classification requires no task-specific training data: the model receives only a carefully designed prompt listing the six target conditions and returns a condition label for each review. This approach tests whether strong general-purpose language understanding can substitute for supervised fine-tuning on this domain.

Zero-shot LLMs provide a complementary perspective to the other two classifiers; rather than learning from label statistics, they apply world knowledge encoded during pre-training. This allows them to reason about drug-condition relationships without ever seeing a labeled example from this dataset. An additional practical motivation is cost: a zero-shot pipeline requires no GPU time for training and can be deployed for under $1 in API costs on this test set.

Approach

Each review was sent to the LLM as a standalone API call. The model was instructed to return only the condition name with no additional commentary, enabling deterministic parsing of the response. Responses that did not exactly match one of the six condition strings were flagged as 'Unknown' and counted as incorrect classifications. To assess the effect of prompt design, a key variable in zero-shot setups, we evaluated two prompt variants on a stratified sample of 3,000 reviews (500 per condition) drawn from the held-out test set:

Prompt Design: Two Variants

Prompt A (Minimal):

Classify the following patient drug review into exactly one of these medical conditions: Birth Control, Depression, Anxiety, Acne, ADHD, Diabetes Type 2.

Review: {review}

Respond with only the condition name, nothing else.

Prompt B (Descriptive):

You are an expert medical text classifier. Identify the condition the patient is being treated for.
Guidelines:

- *Focus on symptom descriptions, drug names, and treatment outcomes mentioned.*
- *Do not rely on the drug name alone; use the full context of the review.*
- *If the review mentions multiple conditions, select the condition that the drug being reviewed is primarily prescribed to treat.*

Choose from: Birth Control, Depression, Anxiety, Acne, ADHD, Diabetes Type 2.
Patient review: {review}
Respond with only the condition name. No explanation.

Prompt A provides the minimal information needed for the task, while Prompt B adds expert framing and explicit reasoning guidance. Comparing the two allows us to quantify how much prompt engineering alone, without any model fine-tuning, can shift classification accuracy.

Evaluation Criteria

Performance evaluation for all three classifiers was done on the test dataset using accuracy and macro precision/recall/F1. Here, the use of macro-averaged criteria is significant for the first two classifiers since it nullifies the advantage of a very popular birth control class while accurately showing the true performance on other minority classes like diabetes type 2 and ADHD.

4. Results and Interpretation of all the classifiers

1. SVM with TF-IDF Performance

In the case of SVM, classification performance on both decomposition techniques was remarkably good. Results were quite similar for both techniques, with OVR being slightly better than ONN (refer to Table 1):

Model	Accuracy	Macro F1	Weighted F1
SVM + TF-IDF (One-vs-Rest)	0.9615	0.9458	0.9614
SVM + TF-IDF (One-vs-One)	0.9614	0.9458	0.9614

Table 1 Overall performance metrics for OvR and OvO SVMs on the held-out test set (n = 18,432).

The differences between OvR and OvO are well below 0.01 percentage points on every metric, indicating that the choice of multi-class decomposition has negligible impact on this task. In what follows, all condition-level analysis uses the OvR model unless otherwise noted. These findings demonstrate that even without contextual word embeddings, a classical TF-IDF + SVM pipeline can extract highly discriminative lexical features from patient drug reviews.

Condition-Level Insights

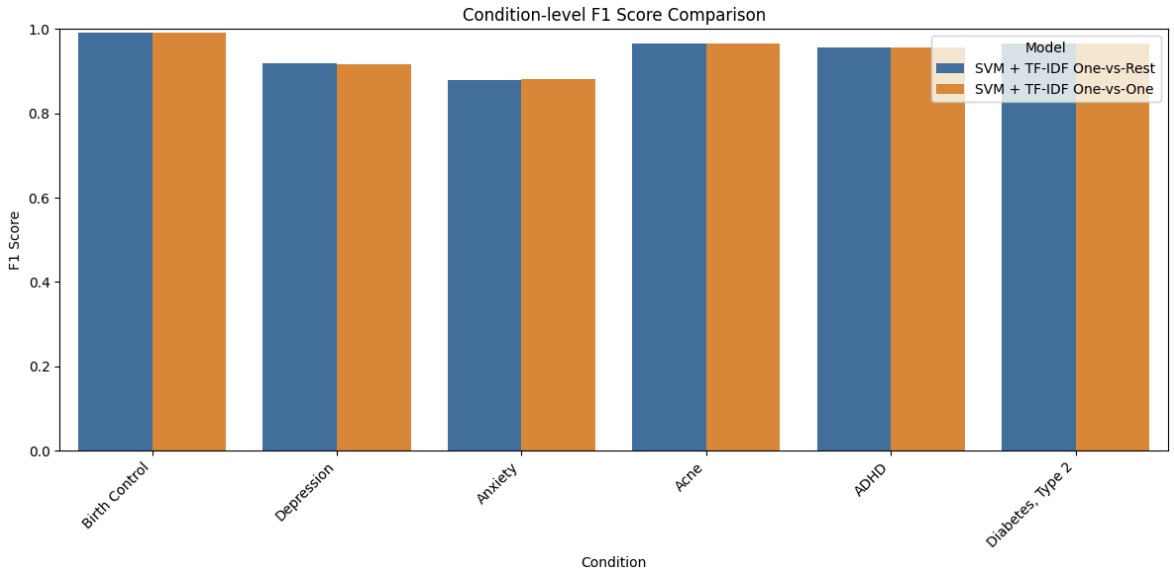


Figure 2. Condition-level F1 scores for OvR and OvO. The two strategies are essentially indistinguishable across all six classes.

There was a significant difference in performance across the different conditions. The condition Birth Control obtained the highest F1 score (≈ 0.99) due to the presence of unique terms such as

“pill,” “period,” "NuvaRing," "Nexplanon," and “spotting.” Similar to birth control, acne (F1 ≈ 0.965) and type 2 diabetes (F1 ≈ 0.965) utilized unique language related to the subject matter. Specifically, Acne’s performance was based on terms associated with dermatology, like "Accutane," “dermatologist,” "Epiduo," and “doxycycline,” while Diabetes’ classification was determined by medical language, including "A1C," “metformin,” "Victoza," "Trulicity," and "Januvia." Interestingly, Diabetes, Type 2 was able to obtain an equally high F1 score despite being the smallest class, demonstrating that term uniqueness is more important than class size when it comes to SVM classification. ADHD (F1 ≈ 0.956) was another well-classified category, aided by language pertaining to stimulant medications and attention deficit disorders, such as “vyvanse,” “adderall,” “concerta,” and “focus.” Depression (F1 ≈ 0.92) performed solidly but more weakly, as its vocabulary overlaps substantially with general affective and psychiatric language. Anxiety (F1 ≈ 0.88) had the lowest F1 of the retained classes.

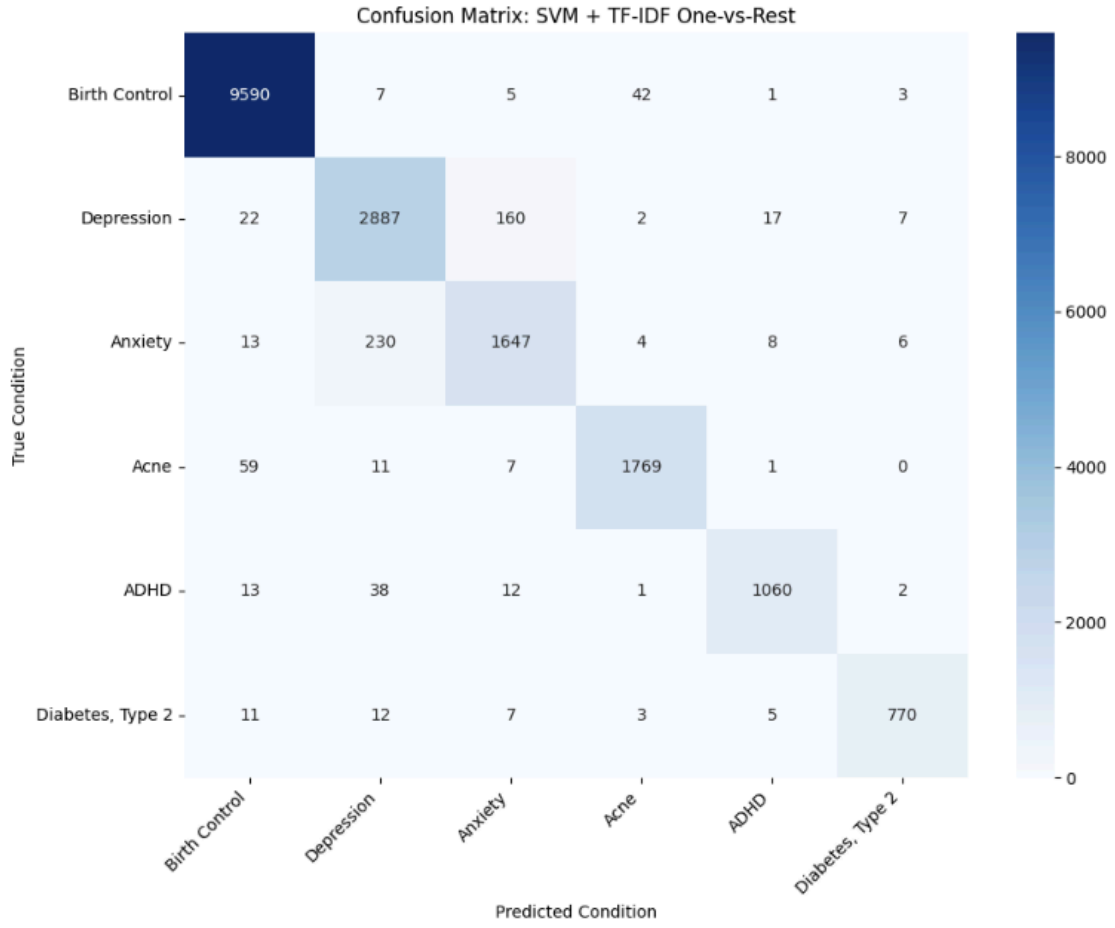


Figure 3. Confusion matrix (One-vs-Rest). Anxiety → Depression is the dominant error pattern.

The confusion matrix identifies the areas that have more errors. The biggest error comes from the box Anxiety to Depression (230 errors, roughly 12% of total cases). This makes up the majority of the loss in recall for anxiety disorders, which is expected given the high similarity between the two. In many cases, the same medication can be used for both conditions, and people who suffer from one will frequently experience symptoms related to the other ("nervousness," "worry," "panic," and "mood"). The second-biggest mistake occurs when there is an error of acne to birth control (59 errors). This error is expected since combined oral contraceptives are often prescribed for acne; although the patient has acne, the reviews can use terms like "birth control."

Condition	Precision	Recall	F1	Support
ADHD	0.97	0.94	0.96	1,126
Acne	0.97	0.96	0.96	1,847
Anxiety	0.90	0.86	0.88	1,908
Birth Control	0.99	0.99	0.99	9,648
Depression	0.91	0.93	0.92	3,095
Diabetes, Type 2	0.98	0.93	0.92	808
Macro Avg	**0.98**	**0.95**	**0.96**	**18,432**

Table 2. Per-Class Performance (SVM + TF-IDF, OvR)

In addition to this confusion pattern, Figure 3 shows that almost all diagnoses are highly clustered around the diagonal, which proves that the SVM + TF-IDF classifier successfully classified a large number of patient reviews for all six diagnoses. Confusion outside of overlapping or similar categories is not as common. For instance, diabetes type 2 and ADHD have quite clear distinctions from each other despite their small dataset sizes, implying that medication terms and symptom words mentioned within the review text could play more important roles than class-unbalanced sparse text in diagnosis. In general, the confusion matrix clearly shows that the SVM + TF-IDF model works best in cases where disorders are characterized by distinctively different lexical patterns.

2. BERT Performance

BERT achieved a macro F1 of 0.94 on the held-out test set (n = 18,432), matching the SVM's macro F1 of 0.9458 within rounding. The training curve was well-behaved: cross-entropy loss fell from 0.5553 in epoch 1 to 0.1837 in epoch 2 and 0.1005 in epoch 3, while training accuracy rose from 81.3% to 94.6% to 97.3%, with no signs of instability or overfitting.

Per-class results on the test set are shown in Table 2. Birth control remained the highest-performing class (F1 = 0.99), consistent with the SVM, reflecting its highly distinctive vocabulary. ADHD (F1 = 0.96), Diabetes, Type 2 (F1 = 0.97), and Acne (F1 = 0.95) all performed strongly. Depression (F1 = 0.92) and anxiety (F1 = 0.88) remained the two weakest classes. Anxiety's recall of 0.89 represents a modest improvement over the SVM, consistent with BERT's ability to use contextual cues to partially disambiguate anxiety from depression, though the boundary remains the dominant challenge across both models.

Condition	Precision	Recall	F1	Support
ADHD	0.96	0.95	0.96	1,126
Acne	0.93	0.97	0.95	1,847
Anxiety	0.88	0.89	0.88	1,908
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Diabetes, Type 2	0.97	0.97	0.97	808
Macro Avg	**0.94**	**0.95**	**0.94**	**18,432**

Table 3. BERT classification report on the held-out test set (n = 18,432).

3. Zero-Shot LLM Performance

The impact of prompt design was substantial. Evaluated on a stratified sample of 3,000 reviews (500 per condition), Prompt A (minimal) achieved a macro F1 of 0.77 and overall accuracy of 77%. Prompt B (descriptive) lifted macro F1 to 0.91 and accuracy to 90%, a 14-point gain from

rewording alone, which is larger than the gap between SVM and BERT. Additionally, Prompt A produced 422 'Unknown' responses where the model output did not match any target condition string; Prompt B reduced this to just 49, demonstrating that clear formatting guidance substantially reduces parse failures.

Model	Accuracy	Macro F1	Unknowns
Zero-Shot LLM (Prompt A)	77%	0.77	422
Zero-Shot LLM (Prompt B)	90%	0.91	49

Table 4. Overall performance metrics for both zero-shot prompt variants on a stratified sample of 3,000 reviews (500 per condition).

Per-Condition Results

Table 5 shows per-condition precision, recall, and F1 for both prompts. The differences across conditions highlight both the difficulty of specific classification boundaries and the sensitivity of zero-shot performance to prompt framing.

Condition	Prompt A			Prompt B		
	Precision	Recall	F1	Precision	Recall	F1
ADHD	0.93	0.95	**0.94**	0.97	0.93	**0.95**
Acne	0.96	0.89	**0.93**	0.98	0.90	**0.94**
Anxiety	0.91	0.78	0.84	0.87	0.81	0.84
Birth Control	0.86	0.98	0.91	0.90	0.97	0.93
Depression	0.81	0.89	0.84	0.82	0.87	0.84
Diabetes T2	1.00	0.10	**0.18**	0.98	0.94	**0.96**
Macro Avg	0.91	0.77	**0.77**	0.92	0.90	**0.91**

Table 5. Per-condition precision, recall, and F1 for both zero-shot prompt variants.

Key Findings

- **Prompt B dramatically outperforms Prompt A on type 2 diabetes:** Prompt A's recall for diabetes collapsed to 0.10 (F1 = 0.18), likely because the model defaulted to other conditions when medical language was ambiguous. Prompt B's instruction to reason from drug-condition context anchored the model to disease-specific terms such as 'metformin,' 'A1C,' and 'glucose,' lifting recall to 0.94 and F1 to 0.96, matching the SVM and BERT on this class.
- **Zero-shot Prompt B (F1 = 0.91) is competitive with fine-tuned BERT (F1 = 0.94) at a fraction of the cost:** BERT required approximately 25 minutes of GPU fine-tuning; Prompt B required no training at all. The 3-point macro F1 gap is meaningful but modest and may be acceptable in settings where labeled data or GPU resources are scarce.
- **Prompt design matters as much as model architecture:** The 14-point macro F1 lift from Prompt A to Prompt B, achieved purely through rewording, exceeds the 2-point gap between SVM and BERT. This finding underscores that for zero-shot classification, engineering the prompt is as critical as selecting the underlying model.
- **An unknown response rate is a practical concern for Prompt A:** The 422 unmatched outputs from Prompt A (versus 49 for Prompt B) indicate that minimal prompts are less reliable in production settings where exact string matching is required. Prompt B's explicit formatting instruction substantially reduced ambiguous outputs.

5. Interpretation

Taken together, the results from all three classifiers reveal consistent and complementary insights about the structure of this classification problem.

What all classifiers agree on:

The anxiety/depression boundary is the dominant challenge across every approach tested. SVM achieves an F1 = 0.88 for anxiety and 0.92 for depression; BERT achieves 0.88 and 0.92, respectively; the zero-shot LLM (Prompt B) achieves 0.84 for both. This convergence across three architecturally distinct classifiers, sparse lexical, contextual transformer, and zero-shot LLM, strongly indicates the difficulty is data-inherent rather than model-specific. Anxiety and depression genuinely share overlapping drug therapies, symptom language, and patient framing

that no classifier fully resolves without additional clinical context such as the drug name or the primary indication. Future work should investigate whether incorporating structured metadata (e.g., the drug field) as an auxiliary feature closes this gap.

Conversely, conditions with class-specific vocabularies, birth control, acne, diabetes type 2, and ADHD are uniformly well-classified across all three approaches, with F1 scores consistently above 0.93. Drug name mentions within the review text, clinical terminologies, and anatomical descriptors act as strong discriminative signals that all feature representations capture effectively. This confirms that term uniqueness is a more important determinant of classification success than class size: Diabetes Type 2, the smallest retained class, achieved F1 $\geq$ 0.96 across all classifiers due to its highly distinctive medical vocabulary.

How the classifiers differ:

Despite broad agreement on which conditions are easy or hard, the three classifiers differ in their accuracy-cost tradeoffs. The SVM with TF-IDF is the strongest supervised baseline: it achieves macro F1 = 0.96, trains in seconds, and is fully interpretable through coefficient inspection. Its performance on lexically distinct conditions is essentially at the ceiling, and it remains the most practical choice when labeled data are available and computational resources are limited.

Fine-tuned BERT achieves macro F1 = 0.94, within 2 points of the SVM, at the cost of approximately 25 minutes of GPU fine-tuning. Its contextual embeddings were expected to provide the greatest advantage on the anxiety/depression boundary, but gains there were modest, suggesting that the semantic overlap between these two conditions exceeds what contextual representations alone can resolve. For the four lexically distinct conditions, BERT and SVM are effectively indistinguishable, confirming that classical sparse features capture nearly all available signal when class vocabularies are well-separated. The weighted cross-entropy strategy successfully preserved minority class performance, with diabetes type 2 achieving F1 = 0.97 despite being the smallest class.

The zero-shot LLM (Prompt B) achieves macro F1 = 0.91 with no training data and no GPU time, making it a remarkably competitive baseline. It trails the SVM by approximately 5 macro

F1 points and BERT by approximately 3 points, a modest penalty for eliminating the need for any labeled examples. A critical finding is that prompt design alone had a larger impact than model architecture: the 14-point macro F1 lift from Prompt A (0.77) to Prompt B (0.91) substantially exceeds the 2-point gap between SVM and BERT. This implies that for zero-shot classification, the prompt is the primary modeling lever and that investment in prompt engineering yields higher returns than investment in architecture selection.

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